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MILITARY SPECIFICATION

CATAPULT, AIRCRAFT EJECTION SEAT: CKU-7/A ASSEMBLY

This specification is approved for use within all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the CKU-7/A aircraft ejection seat catapult assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-14814	Cartridge, Impulse M270
MIL-I-46625	Initiator, Cartridge Actuated: M3A2 and M3A1C Assembly

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC: N4413 FSC 1377

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MIL-STD-1167	Ammunition Data Card
MIL-STD-1168	Ammunition Lot Numbering
DOD-STD-2101	Classification of Characteristics
MS 17983	Compass-Magnetic, Pilots Standby
MS 28741	Hose Assembly—Detachable End Fitting, Medium Pressure

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

NAVAL ORDNANCE STATION, INDIAN HEAD, MD (CODE IDENT 14083)

11820361	Catapult, Aircraft Ejection Seat, Assembly

NAVAL SEA SYSTEMS COMMAND (CODE IDENT 53711)

5184318	Box, Packing Assembly for Rocket Catapult

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subject to first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials, parts and assemblies shall be in accordance with dwg 11820361, this specification and the documents referenced herein.

3.3 Age.

3.3.1 Explosive components. All explosive components, including but not limited to ignition powders and igniters, shall have been manufactured or recertified within 12 months prior to the contract delivery date for the assembly lot. Primers shall be manufactured and certified within 24 months or recertified within 12 months prior to the contract scheduled delivery date for the assembly lot.

3.3.2 Propellant. The propellant shall be manufactured within 150 days prior to the contract scheduled delivery date for the catapult assembly lot.

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3.3.2.1 Propellant morphology. Two propellant sleeve and grain assemblies submitted with the first article sample shall be examined in accordance with 4.4.3 for presence of oxidizer agglomerates, binder rich areas, uncured binder or other anomalies in propellant morphology.

3.4 Ballistics. The catapult, loaded with acceptable explosive components shall meet the following ballistic performance characteristics and function without mechanical failure (see 6.5 for definition of terms).

3.4.1 Booster section. The booster section shall comply with the requirements of table I using an ejected mass of 459 lbm.

3.4.2 Rocket section. The rocket section shall comply with the requirement of table II with the nozzle angle set at 40°F.

TABLE I. Booster requirements.[1]

Temperature (°F)	−65 to −70° F	65 to 70° F	160 to 165° F
Acceleration (g), max	20	18	20
Rate change of acceleration (g/s), max	300	300	300
Axial thrust (lbf), max	7000	7000	7300
Separation velocity (ft/s), minimum	37	40	42
Launch time (ms), max	210	200	200
Ignition delay (ms), max	30	30	30

[1]See 6.5 for definition.

TABLE II. Rocket requirements.[1]

Temperature (°F)	−65 to −70° F	65 to 70° F	160 to 165° F
Resultant impulse (lbf/s), min	1100	1100	1100
Resultant thrust (lbf), max	7350	7350	7600
Ignition delay (ms), max	25	25	25
Action time (ms), max	550	450	450

[1]See 6.5 for definition.

3.4.3 Initiation. The rocket catapult shall fire when initiated by a source delivering an output equal to that of an M3A1 initiator in accordance with MIL-I-46625 at the end of 10 feet of aircraft hose size No. 4 of MS 28741 at 70° ± 5°F. Equivalent initiation input for test is 1100 psi pressure source producing a 40,000 to 50,000 psi per second positive rate of change.

3.4.4 Other requirements. In addition to meeting the ballistic requirements, none of the following shall occur during test firing of the rocket catapult:

a. No unit component including but not limited to O-rings, springs, or pieces of propellant shall be ejected
b. No component subject to loading shall fail
c. No excessive nozzle erosion, hot spots, or burn-through shall occur.

3.5 Torque. The components specified in table III shall show no loosening or deformation upon application of breakaway torque.

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TABLE III. Torque values.

Components	Seating or bottoming (inch lbf)	Breakaway (inch lbf)
Booster tube assembly 10551422 to head 10551412	400 + 50, −0	200
Launcher tube assembly 10551423 to mount 10551413	400 + 50, −0	200

3.6 Residual magnetism. The assembled catapult shall not deflect the indicator of a compass in accordance with MS 17983 more than 5° in either direction when passed in front of the compass at a distance of 6 inches.

3.7 Nozzle integrity. Nozzle ball/retainer O-ring shall not show any leakage throughout the operational range of the swivel nozzle when pressurized to 40 psig. Application of 40 psig internal pressure to the sealed nozzle/retainer/sleeve assembly shall cause the nozzle to rotate fully.

3.8 Release O-ring integrity. Release O-rings shall show no indications of leakage when the launcher tube-mount subassembly is pressurized to 200 psig.

3.9 Nozzle/sleeve actuation force. The force required to move the sleeve an adequate distance to cause full rotation of the nozzle in a complete nozzle/retainer/sleeve assembly, shall not exceed 25 lbf.

3.10 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the components regardless of contour configuration.

3.11 Radiographic requirements. Each loaded rocket catapult shall be radiographically inspected for the presence, condition, and correct placement of internal components.

3.12 Environmental requirements. The rocket catapult assembly shall withstand the environmental conditioning requirements of 3.12.1 through 3.12.5. The rocket catapult assemblies subjected to this conditioning shall show no evidence of disintegration of components when examined in accordance with 4.5.5.2 and shall meet the ballistic requirements of 3.4 when test fired in accordance with 4.4.2.

3.12.1 High temperature. The rocket catapult shall withstand the high temperature test of 4.4.1.1.

3.12.2 Low temperature. The rocket catapult shall withstand the low temperature test of 4.4.1.2.

3.12.3 Temperature shock. The rocket catapult shall withstand the temperature shock test of 4.4.1.3.

3.12.4 Vibration. The rocket catapult shall withstand the vibration test of 4.4.1.4.

3.13 Identification marking. Each rocket catapult shall be identified in accordance with dwg 11820361, to include an identification label, a warning label, and an installation label.

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3.13.1 Production lot designations. The production lot number shall follow MIL-STD-1168. The construction of the lot shall be as follows:

IHM	75	D	002	007
a	b	c	d	e

where: a = Manufacture's Identification Code

b = The year of manufacture of the oldest batch of propellant in the lot.

c = The month of manufacture of the oldest batch of propellant in the lot expressed as an alpha code in accordance with 4.1.3 of MIL-STD-1168

d = Lot interfix number (controlled by Naval Ordnance Station, Indian Head, Maryland 20640-5000 or contracting activity)

e = Lot sequence number—lot number (manufacturer is responsible for the assignment of lot numbers).

3.14 Workmanship. The workmanship shall be that required by the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein and on the applicable drawings and specifications. All components and assemblies shall be free from burrs, contamination, corrosion, sharp edges, or foreign material which could result in malfunction of the unit or components, or be a safety hazard in handling.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65° to 95°F
b. Altitude: Normal ground level and atmospheric pressure
c. Vibration: None
d. Humidity: Room ambient to 95 percent relative, maximum

4.3.1 Inspection and test equipment. Inspection and test equipment accuracies shall be as listed in Table IV.

TABLE IV. Test equipment accuracies.

Parameter	Equipment accuracy
Maximum thrust	±2.0 percent
Time	±0.001 second
Maximum acceleration	±5.0 percent

4.4 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2), a first article sample of 21 loaded rocket catapult assemblies, manufactured in accordance with 3.1, shall be submitted to the Naval Ordnance Station, Indian Head, MD 20640-5000 for first article inspection. The first article samples shall be subjected to the quality conformance inspection of 4.5 with the exception of the test firing of 4.5.5, followed by environmental testing, radiographic examinations,and static firing in the sequence specified on table V. Acceptance of the first article sample shall be based on no defects in the sample.

TABLE V. First article evaluation test program.[1]

Test Para.	Test	Unit number																			
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
4.5.5.2	Radiographic inspection	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a	a
4.4.1.4	Vibration 165°F	b									b		b									
4.4.1.4	Vibration −65°F		b	b								c										
4.4.1.3	Temperature shock				b	b	b				c	b	c									
4.4.1.1	High temperature							b	b		e	d	e									
4.4.1.2	Low temperature									b	d	e	d									
4.5.5.2	Radiographic inspection	c	c	c	c	c	c	c	c	c	f	f	f									
4.4.2	Static fire 165°F	d				d	d		d		g		g				b	b	b			
4.4.2	Static fire 70°F													b	b	b						
4.4.2	Static fire −65°F		d	d	d			d		d		g								b	b	b

[1]Letters denote sequence of testing.

4.4.1 Environmental tests.

4.4.1.1 High temperature. The rocket catapult assembly shall be subjected to a high temperature test in accordance with Procedures I of Method 501 of MIL-STD-810 with the exception that the test item shall be maintained at 165°F for 50 hours minimum.

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4.4.1.2 Low temperature. The rocket catapult assembly shall be subjected to a low temperature test in accordance with Procedure I of Method 502 of MIL-STD-810 with the exception that the test item shall be maintained at −65°F for 50 hours minimum.

4.4.1.3 Temperature shock. The rocket catapult assembly shall be subjected to a temperature shock test in accordance with Procedure I of Method 503 of MIL-STD-810 with the exception that the conditioning time at each temperature shall be 8 hours.

4.4.1.4 Vibration. The rocket catapult assembly shall be subjected to random vibration for jet aircraft in accordance with Method 514 of MIL-STD-810. Before vibration, units shall be temperature conditioned in accordance with table VI. The input vibratory acceleration factor shall be 0.25 G2/HZ. If the rocket catapult must be removed from the environmental conditioning chamber or if the temperature falls outside the specified temperature limits for a period of more than 5 minutes, the testing shall be stopped and the unit shall be reconditioned. A reconditioning time of 2 minutes for every minute the unit is outside the specified temperature envelope shall be required. A continuous temperature history of the unit shall be kept for all conditioning and testing time. Unless otherwise noted, the rocket catapult shall be mounted on a test fixture in a manner that will simulate actual seat/aircraft installation and excited through the attachment fittings.

TABLE VI. Static firing sample size and temperature conditioning plan.

Lot size	Sample size −65°F to −70°F	Sample size 160°F to 165°F	Total firing sample	Acceptability constant K[1]
Up to 40	3	3	6	—
41 — 65	4	4	8	—
66 — 110	5	5	10	—
111 — 180	7	7	14	—
181 — 300	10	10	20	—
301 — 340	11	11	22	1.41
341 — 380	12	12	24	1.43
381 — 420	13	13	26	1.45
421 — 500	15	15	30	1.47

[1]The acceptability constant (K) for a lot size up to 300 loaded units shall be derived in accordance with MIL-STD-414. K for a lot size between 301 and 500 loaded units shall be as indicated.

4.4.2 Static firing test. Within 30 days after completion of environmental tests, the rocket catapults shall be static fired at an activity designated by the contracting activity (see 6.2). The sample shall be temperature conditioned per table VI for a minimum of 8 hours and test fired within 10 minutes after removal from the conditioning chamber to verify conformance to the requirements of 3.4. Nonconformance with table I, or any other requirement of 3.4.1 shall be cause for rejection of the first.

4.4.3 Propellant morphology. Propellant from the sleeve and grain assemblies shall be subjected to a morphological examination. This examination shall include, but is not limited to microscopy and chromatography. Test results will be used to document propellant homogeniety, ammonium perchlorate particle size distribution, and presence of oxidizer agglomerates, binder rich areas, uncured binder, or other anomalies in propellant morphology. Findings will be used for information only to provide baseline data for future reference.

4.5 Quality conformance inspection. The quality conformance inspection shall consist of the examinations and tests of 4.5.4 thru 4.5.5.7 conducted on samples selected as specified in 4.5.2. The contractor is responsible for assuring that all rocket catapults offered for acceptance meet all the requirements of the applicable drawings and specifications. Items of inspection shall be serialized and inspection data traceable to a specific serial number.

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4.5.1 Test reports. When specified in the contract or purchase order (see 6.2.2), the contractor shall furnish test reports to the contracting activity. Test reports shall include the following information:

a. Propellant description (see 6.4.1)
b. Static firing performance test results (see 4.5.5.1 and 6.4.2)
c. Summary sheet (see 6.4.3)
d. Ammunition data cards (see 5.2)
e. Propellant/assembly production records (see 6.4.5)
f. Certification of age of explosive components and propellant (see 3.3).

4.5.2 Sampling.

4.5.2.1 Lot. A lot shall consist of an identifiable group of assembled units produced by a homogeneous process and loaded with a single lot of acceptable cartridges in accordance with MIL-C-14814 (see table VI).

4.5.2.2 Sampling for examination. Designation of defects (critical, major and minor) shall be in accordance with DOD-STD-2101.

a. Critical and major defects: Each rocket catapult assembly shall be examined for all critical and major defects.

b. Minor defects: Samples for examination shall be selected in accordance with inspection level II of MIL-STD-105.

4.5.2.3 Sampling for tests. Sampling for tests shall be in accordance with the following:

4.5.2.3.1 Static firing tests. Rocket catapult assemblies from each lot shall be submitted to the Naval Ordnance Station, Indian Head, Maryland 20640-5000 (see 6.2) for static firing in accordance with 4.5.5.1. The sample size shall be in accordance with table V. Sample rocket catapult assemblies shall not be preselected during production. Rocket catapult assemblies selected for static firing shall not be included as a part of the quantity specified for delivery in the contract (see 6.2). The ballistic sample size shall be divided as equally as possible among rocket motor batches. Each batch of propellant must be represented by at least one sample per temperature extreme.

4.5.2.3.2 Breakaway torque. Each rocket catapult assembly shall be subjected to the breakaway torque test of 4.5.5.3.

4.5.2.3.3 Residual magnetism. Each rocket catapult assembly shall be subjected to the residual magnetism test of 4.5.5.4.

4.5.2.3.4 Nozzle O-ring integrity. Each rocket catapult assembly shall be subjected to the nozzle ball/retainer O-ring leak test of 4.5.5.5.

4.5.2.3.5 Release O-ring integrity. Each rocket catapult assembly shall be subjected to the release O-ring leak test of 4.5.5.6.

4.5.2.3.6 Nozzle/sleeve actuation force. Each rocket catapult assembly shall be subjected to the nozzle/sleeve actuation force test of 4.5.5.7.

4.5.3 Resubmission of rejected lot. A sample of rocket catapult assemblies shall be selected in accordance with 4.5.2.3.1 and submitted when, with prior approval of the contracting activity, the contractor requests that the rejected lot be test fired under different conditions, or the contractor believes a rejected lot may be reworked to meet the requirements of this specification.

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4.5.4 Examination.

4.5.4.1 Critical and major defects. Each rocket catapult assembly shall be examined for critical and major defects specified in 4.5.2.2a and the applicable drawings. Any rocket catapult assembly found to contain a critical or major defect shall be rejected.

4.5.4.2 Minor defects. Sample rocket catapult assemblies selected in accordance with 4.5.2.2b shall be examined for minor defects in accordance with the applicable drawings and 3.13. The Acceptance Quality Level (AQL) shall be 2.5 defects per hundred units.

4 5.5 Tests. All test and performance requirements are classified as major characteristics. Sampling shall be as specified by 4.5.2.3.

4.5.5.1 Static firing test. Rocket catapults selected in accordance with 4.5.2.3.1 shall be static fired at the Naval Ordnance Station, Indian Head, Maryland 20640-5000 (see 6.2) within 30 days after submission for acceptance of the rocket catapult lot. The sample catapult shall be subjected to the temperature conditions specified in table VI for a minimum of 8 hours and static fired within 10 minutes after removal from the conditioning chamber.

4.5.5.1.1 Method of testing. The rocket catapult shall be initiated by an M3A1 initiator or simulated method to verify compliance with 3.4 and tables I and II. The head of the rocket catapult shall be positioned horizontally with the head statically supported and the 459 lbm mass attached to the mounting lugs and permitted to move freely. Performance of the booster section shall be obtained using a minimum 25 KHz accelerometer mounted on the mass. Maximum accuracies shall be as shown in table IV. The ballistic data shall be analyzed and accepted in accordance with MIL-STD-414, standard deviation method, single specification limit, variability unknown with an AQL of 2.5 percent.

4.5.5.1.2 Post static-firing disassembly. A post static-firing disassembly and inspection shall be conducted on all first article and lot acceptance test units. Detailed observations on conditions, including marginal conditions of individual components shall be noted. All findings shall be evaluated against the requirements of 3.4.4. When specified in the contract or purchase order (see 6.2), a disassembly findings report shall be submitted to the contracting activity.

4.5.5.2 Radiographic inspection. Rocket catapults shall be radiographically inspected in accordance with MIL-STD-453, radiographic quality level 1. Any deficiency shall be cause for rejection of the unit. Radiographs shall be capable of detecting defects and rocket catapult components 0.03 inch in length or width, whichever is smaller. If all components cannot be shown using standard radiographic techniques, neutron radiography shall be employed to show these components. When specified in the contract (see 6.2), all radiographs taken of the rocket catapult assembly or components thereof shall be forwarded to the activity designated by the contracting activity. Otherwise, all radiographs shall be inspected on-site by a designated Government representative. Acceptance of the loaded rocket catapult lot shall be subject to approval of the radiographs by the designated Government activity.

4.5.5.3 Breakaway torque. After ascertaining that each component specified in table III has been seated with 400 + 50, −0 inch-pounds of torque, a breakaway torque of 200 inch-pounds shall be applied. Any evidence of loosening or deformation of any component shall result in rejection of the catapult assembly.

4.5.5.4 Residual magnetism. Each catapult assembly shall be slowly passed 6 inches in front of an MS 17983 compass along the entire length of the catapult in accordance with figure 1. If the compass indicator deflects more than 5° in either direction, the catapult assembly shall be rejected. The test shall be performed in an area free from local magnetic affects.

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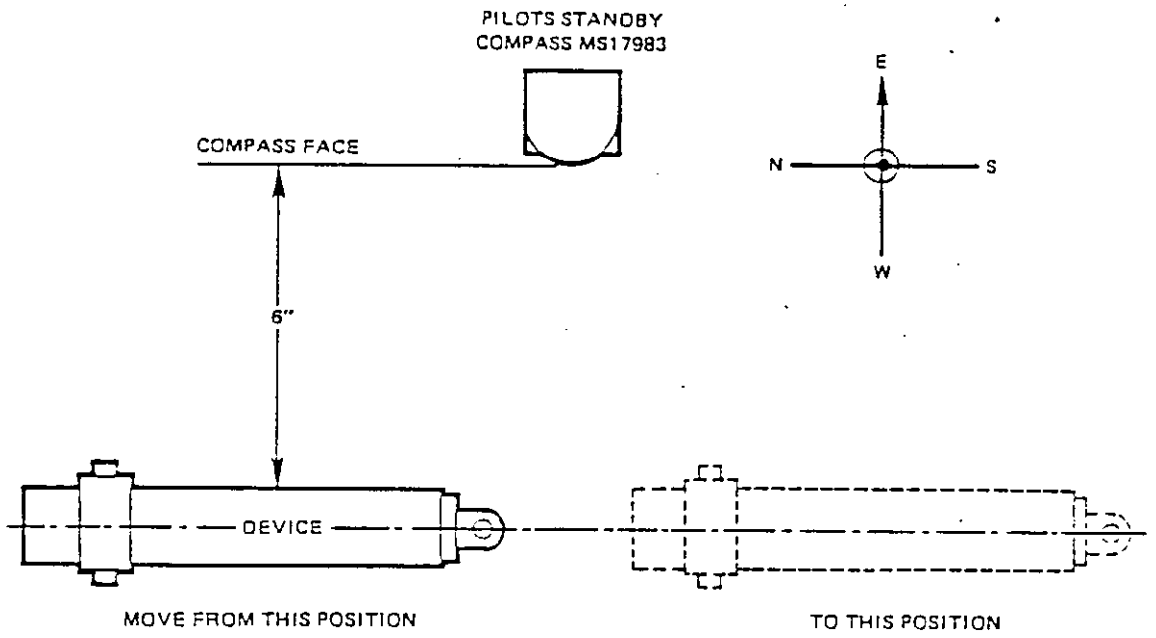


FIGURE 1. Layout for determination of residual magnetism in device.

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4.5.5.5 Nozzle O-ring integrity. The retainer-nozzle ball O-ring shall be tested for leakage by partially assembling the catapult so that the retainer nozzle subassembly and motor tube are assembled. The assembly shall then be pressurized to 40 psig and the nozzle swiveled through its operational range. Any loss of pressure within 15 seconds after nozzle has been fully swiveled shall result in rejection of the catapult assembly. Application of 40 psig internal pressure to the sealed assembly must cause the nozzle to rotate from the axial position to the final canted position without application of any external force. Failure to rotate completely shall result in rejection of the assembly.

4.5.5.6 Release O-ring integrity. The release O-rings shall be tested for leakage by pressurizing the launcher tube-mount subassembly with nitrogen at 200 $\pm$ 5 psig. Any loss of pressure within 15 seconds of pressurization shall result in rejection of the catapult assembly.

4.5.5.7 Nozzle/sleeve actuation force. The retainer/nozzle/sleeve subassembly shall be tested to permit measurement of the axial force which is necessary to move the sleeve an adequate distance to cause full rotation of the nozzle. The assembly shall include all internal O-rings when tested. Force measurement shall be performed with a suitable force gage accurate to 1 pound. Measurement of an axial force exceeding 25 pounds shall result in rejection of the assembly.

4.5.6 Packaging inspection. The packaging, packing, and marking shall be inspected to verify conformance with the requirements of section 5.

5. PACKAGING

5.1 Preservation, packaging, packing and marking. Unless otherwise specified in the contract (see 6.2), preservation, packaging, packing, and marking shall be in compliance with dwg 5184318.

5.2 Data cards. Data cards (DD form 1650) should be completed for each loaded rocket catapult lot. Copies should be provided in accordance with contract requirements and one copy should be forwarded to the Commanding Officer, Naval Ordnance Station, (Code 5120), Indian Head, MD 20640-5000.

6. NOTES

6.1 Intended use. The CKU-7/A rocket catapult assembly is intended to propel an aircraft ejection seat and its occupant safely out and away from a jet aircraft.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Inspection conditions, if other than specified (see 4.3)
c. Quantity required (exclusive of first article and static firing test samples) (see 4.4 and 4.5.2.3.1)
d. When a first article sample is not required (see 3.1, 4.4, and 6.6)
e. Assigned activity for first article inspection and static firing tests (see 4.4, 4.4.2, 4.5.2.3.1, 4.5.5.1 and 6.7)
f. Whether test reports are required (4.5.1 and 6.3)
g. If radiographs are to be forwarded (4.5.5.2)
h. Packaging requirements if other than level A (see 5.1)
i. Safety precautions (see 6.8)
j. Disposition of hardware, scrap propellant and metal parts (see 6.9)
k. That the contractor retain radiographs on file (see 6.10).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DoD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

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Paragraph No.	Data requirement title	Applicable DID No.	Option
4.5.1	Reports, Test	DI-T-2072	Contractor Format
5.2	Ammunition Data Card	DI-MISC-80043	

(Data Item Descriptions related to this specification, and identified in section 6, will be approved and listed as such in DoD 5010.12-L, AMSDL. Copies of Data Item Descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the items should be a first article sample. The first article should consist of 21 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Description of data. The following paragraphs are a description of data to be included in test reports per 4.5.1 when specified in the contract or order (see 6.2.2).

6.4.1 Propellant description. The following propellant information should be submitted with each production lot. The materials and components of each propellant batch should be listed with the manufacturer, lot number, and date of manufacture. The burning rates of 2500, 5000, and 7500 psi performed at −60°, 70° and 165°F for each pressure (three samples at each temperature and pressure per mix). The results of tensile strength and elongation tests at −65°, 70° and 165°F with a pull rate of 2 inches/minute (three samples at each temperature per mix), heat of explosion (3 samples per mix). Verification that ingredients were properly weighed and added to each mix (mix sheet(s)).

6.4.2 Static firing performance tests. All computations and data required for ballistics evaluation, generated during quality conformance inspection static firings per 4.5.5.1 should be reported together with firing traces and the following information:

a. Grain designation, manufacturer, lot number, and date of propellant manufacture
b. Type of test equipment
c. Deviations from standard testing procedures
d. Additional comments (malfunctions, unusual occurrences)
e. Disassembly findings report (see 4.5.5.1.2).

6.4.3 Summary sheet. A summary of the results of the tests for the parameters listed in table I should be submitted to the designated Government activity for each inspection lot. The summary should include the sample size, number and nature of defects found, and the disposition (accepted or rejected).

6.4.4 Propellant/assembly production records. The contractor should prepare and retain, for the service life of the unit propellant, production records including batch number, mix designation, rocket motor lot number, rocket motor serial number, cartridge lot number, and cartridge serial number. These should be cross referenced to the assembled rocket lot number and to specific rocket catapult serial numbers.

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6.5 Definitions.

6.5.1 Mechanical failure. A mechanical failure is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5.2 Booster ignition delay. Booster ignition delay is defined as the time that elapses between the firing of the catapult primer (TO) and the beginning of the continuous rise (T1) on the booster thrust-time record (see figure 2).

6.5.3 Launch time. Launch time is defined as the time between the beginning of the continuous rise (T1) and sudden drop (T2) on the booster thrust-time record (see figure 2).

6.5.4 Separation velocity. Separation velocity is defined as the average velocity of the simulated ejected mass propelled horizontally at catapult separation measured over the interval from separation to 12 inches after separation.

6.5.5 Rate of change of acceleration. Rate of change of acceleration is defined as the maximum positive slope of the acceleration time record measured over any 30 milliseconds (ms) between the beginning of the continuous rise and the maximum acceleration on the acceleration time record (see figure 2).

6.5.6 Rocket ignition delay. "Rocket ignition delay" is defined as the time between the sudden drop (T2) on the booster-thrust-time record and the point at which the thrust on the rocket axial thrust record reached 10% of its maximum value (see figure 2).

6.5.7 Action time. Action time is defined as the time between the points where the rocket axial thrust reaches 10% of its maximum value at the start of burning and falls below 10% of the maximum value at the end of burning (T4) (see figure 2).

6.5.8 Rocket motor resultant thrust. Rocket motor resultant thrust is obtained by vectorially adding the two maximum axial and normal components of rocket thrust that occur at one time.

6.5.9 Thrust-time integral. Thrust-time integral is defined as the integral of the area under the rocket thrust-time bounded by the action time limits (T3) to (T4) (see figure 2).

6.5.10 Rocket resultant impulse. Rocket resultant impulse is defined as the resultant of the integrals of the areas under the rocket axial thrust-time and normal thrust-time records bounded by the action time limits (see figure 2).

6.5.11 Propellant batch. Propellant batch is a mix in which each ingredient is composed entirely of one homogeneous lot.

6.5.12 Rocket catapult lot. Rocket catapult lot is a group of units composed of one or more rocket motor propellant batches, one lot of metal parts, and one lot of catapult cartridges.

6.6 First article sample waiver. First article samples submitted and approved on a recent contract (a contract remaining in production or a contract out of production for less than one year) may be accepted by the contracting activity in lieu of an additional first article inspection. When the first article sample is waived (see 6.2), the acquisition document should contain a statement specifying that the standards of workmanship exhibited by the previously approved first article sample shall determine the minimum requirements of the current contract or order.

6.7 Test facility for first article inspection. The Naval Ordnance Station, Indian Head, Maryland 20640-5000 is the Design Agent for the product covered by this specification. As such they are equipped to perform the first article inspection and static firing tests specified herein. Except when special conditions warrant otherwise, the Naval Ordnance Station, Indian Head, Maryland 20640-5000 should be designated as the test facility for the first article inspection and the static firing tests (see 6.2).

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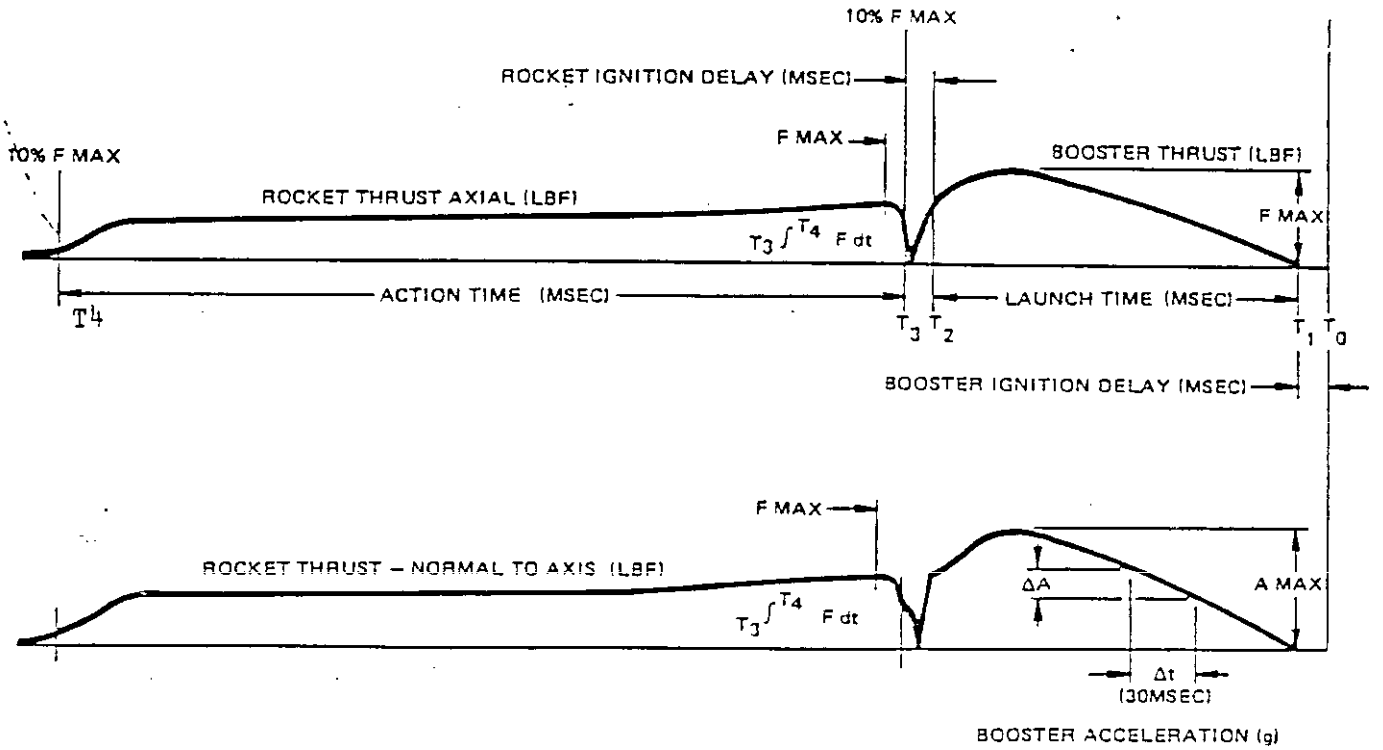


FIGURE 2. Performance curve.

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6.3 Safety precautions. The safety precaution requirement of the "DOD Contractors" Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract or order as required by DOD Federal Acquisition Regulation (FAR) Supplement Part 28, Sub-Part 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP 5) should be made applicable.

6.3.1 Explosive hazard classification. The explosive hazard classification for this rocket catapult has been designated as:

Explosive Hazard Classification—Class 2
Storage Compatibility Group—Group J
Department of Transportation (DOT) Hazard Class—E
DOT Marking—Propellant Explosive

6.9 Disposition of hardware, scrap propellant, and metal parts. The contract should specify that all items purchased become the property of the U.S. Government; no hardware, propellant, or metal parts acceptable, scrapped, or tested shall be disposed of without authorization of the contracting activity; all items shall be made available to the contracting activity upon request; and that the disposal of any tested hardware, propellant, or metal parts without authorization by the contracting activity will result in the test being declared invalid, and a retest will be required.

6.10 Retention of radiographs. The contract should specify that all radiographs of accepted lots of rocket catapults be retained on file by the contractor for 8 years or 1 year past the service life of the unit, whichever is longer.

6.11 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to previous issue due to the extensiveness of the changes.

6.12 Subject terms (keyword) listing.

Booster
Explosive
Nozzle
Propellant

Custodians:
Navy—OS
Air Force—99

Review:
Air Force—70

Preparing activity:
Navy—OS
(Project 1377 — 0908)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48568B	CATAPULT, AIRCRAFT EJECTION SEAT: CKU-7/U ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	7b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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